Kunal Chevuri

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Education

Prosper High School — Prosper, TX
Expected graduation: May 2027 — 08/2023 – 05/2027

- **Grade point average:** 5.363 of 6.0 weighted (4.0 unweighted) | **Class rank:** 15 of 637 | **SAT:** 1540, December 2025 (Mathematics 800, Evidence-Based Reading and Writing 740)
- **Advanced Placement examination scores, May 2026:** Calculus BC: 4 | Statistics: 4 | Physics 1: 4 | Microeconomics: 4 | United States History: 4
- **Advanced Placement examination scores, May 2025:** Precalculus: 5 | Computer Science A: 5 | Computer Science Principles: 4 | World History: 4. **May 2024:** Human Geography: 5
- **Current coursework (2026–2027):** Advanced Placement Physics C (Mechanics and Electricity and Magnetism), Psychology, United States Government and Politics, and Macroeconomics; dual-credit Business Mathematics and Business Calculus through Collin College
- **Computer science readiness:** Advanced Placement Computer Science A (Java), score of 5, extended well beyond the course through the research software listed below, written in Python and published across six public repositories on GitHub
- **Academic recognition:** National Merit Commended Scholar (August 2026); Advanced Placement Scholar with Distinction (July 2025); National Honor Society, inducted Fall 2025

Research Publications

Peer-reviewed research conducted with faculty advisors at three universities and a national neuroscience research program. The Institute of Electrical and Electronics Engineers (IEEE) is the largest professional body in computing and engineering, and its conferences are peer reviewed; workshop and symposium papers appear in the same conference proceedings as main-track papers. Author's name shown in **K. Chevuri**; venue names link to the conference, workshop or journal. Numbering is referenced throughout this resume.

Published / Accepted for Publication

[1] **K. Chevuri**, "Visual Degradation Breaks a Fine-Tuned Vision-Language Model but Not a Frozen Self-Supervised Encoder: A Robustness Study for Roadway Flood Monitoring," in *Proceedings of the IEEE Massachusetts Institute of Technology Undergraduate Research Technology Conference*, Cambridge, MA, October 9–11, 2026. Code
Sole-authored. Under combined blur, darkness and noise a fine-tuned vision-language model missed every severe flood, while a frozen I-JEPA encoder held steady under noise.

[2] **K. Chevuri**, A. Peredo, and S. Chuprov, "Diagnosing the Synthetic-to-Real Gap in Small VLMs for Deployable Flood Monitoring," accepted to the *IEEE International Conference on Cognitive Machine Intelligence*, San Jose, CA, November 2026. Code | Dataset (Zenodo)
Generated 1,005 synthetic images from real traffic-camera frames; vision-language models fine-tuned on them transferred poorly to real photographs, while frozen I-JEPA reached 80.8% macro-recall.

[3] **K. Chevuri**, A. Poon, R. Mahabir, W. Mgomezulu, M. Komwa, and O. Gkountouna, "An Open Deal-Level Dataset of African Agrifood Investment, 2015–2025, and the Reproducible Pipeline That Builds It," accepted to the *High School Symposium* at the *IEEE International Conference on Big Data*, Phoenix, AZ, December 14–17, 2026. Joint first author with A. Poon. Code & Data
Publishes a free database of 1,060 agricultural investment deals across 389 transactions, 313 companies and 30 countries, drawn from 121 named sources, together with the software that rebuilds it identically on every run so other researchers can verify it.

[4] **K. Chevuri** and S. Chuprov, "Masking or Noise? Null-Calibrated Auditing of Per-Dimension Failures Behind Composite Wellness Scores in Federated Learning," accepted to the *ARRL-Health Workshop* at the *IEEE International Conference on Bioinformatics and Biomedicine*, Dallas, TX, December 1–4, 2026. Code
Introduces the Masking Gap, a measure of how far a single combined health score hides a person's worst individual health problem, tested on 8,353 records from the National Health and Nutrition Examination Survey.

[5] **K. Chevuri**, L. Ileto, and M. Aledhari, "Auditing Data Leakage and Reporting Gaps in EEG Connectivity Deep Learning for Depression Detection," accepted to the *7th International Workshop on Machine Learning for EEG*

Signal Processing at the *IEEE International Conference on Bioinformatics and Biomedicine*, Dallas, TX, December 1–4, 2026.

Identifies five ways published brain-wave studies let test data leak into training, then audits 18 depression-detection papers against them.

Submitted / Under Review

[6] **K. Chevuri** and S. Chuprov, "Robustness or Weighting? Re-Evaluating Federated Aggregation on Heterogeneous Health Data," submitted to the *2nd Workshop on Artificial Intelligence in Big Data Analytics for Healthcare* at the *IEEE International Conference on Big Data*, Phoenix, AZ, December 14–17, 2026. Code

624 experiments on 382,709 United States Centers for Disease Control and Prevention health records show that a simple averaging baseline recovers 79 to 98% of the advantage claimed by far more complex published methods.

[7] **K. Chevuri**, A. Sheikh, and M. Aledhari, "Measuring Evasion in Institutional Question-Answering: A Training-Free LLM-as-Judge Framework with Cross-Domain Validation," submitted to the *3rd IEEE International Workshop on Large Language Models for Finance* at the *IEEE International Conference on Big Data*, Phoenix, AZ, December 14–17, 2026. Code

Uses a large language model to score how evasively corporate executives answer analyst questions, across 3,350 exchanges from earnings calls filed with the United States Securities and Exchange Commission, checked against two human raters and two outside datasets.

[8] **K. Chevuri**, A. Poon, R. Mahabir, W. Mgomezulu, M. Komwa, and O. Gkountouna, "Do Development Finance Institutions Crowd In Private Capital? Network Evidence from African Agrifood Investment, 2015–2025," submitted to *World Development*, 2026. Joint first author with A. Poon.

Economics journal submission asking whether public development banks genuinely attract private investors into African agriculture, answered by mapping who invests alongside whom across 420 investors.

[9] **K. Chevuri** and M. Aledhari, "Predicting U.S. Public-Company Failure from Free Regulatory Filings: A Reproducible Pipeline, an Operating-Point Analysis, and an Audit of Its Own Failure Modes," submitted to *Expert Systems with Applications*, 2026. First author.

Predicts which public companies will fail, using 69 financial measures across a 57,000 company-year dataset built entirely from free government filings, outperforming the standard Altman Z-score benchmark by 71%.

[10] **K. Chevuri** and M. Aledhari, "Decomposing the Post-2020 Rise in the Cancer Multiple-Cause to Underlying-Cause Ratio: COVID-Involved and Non-COVID Components, United States 2003–2024," submitted to *Demographic Research*, 2026. First author.

Separates how much of the post-2020 rise in United States cancer mortality reporting is attributable to COVID-19 and how much is not, across 60 million death records.

Conference Presentations & Posters

[11] Paper [6] presented at *University of North Texas Research Day*, Denton, TX, October 1, 2026, and at the *National Research Conference* at the University of Pennsylvania, October 2–3, 2026.

[12] Paper [10] presented at the *Texas Junior Academy of Science*, October 8, 2026.

[13] **K. Chevuri**, L. Ileto, C. Song, A. Zaitkhan, M. Mishra, S. Mena, R. Claret, R. Dabas, S. Chun, J. Sao, and P. Maheshwari, "Deep Learning and EEG Functional Connectivity for Major Depressive Disorder: A Bibliometric Analysis of Research Trends and Validation Practices," electronic poster, ThinkNeuro Cognitive and Computational Neuroscience Program, 2026.

First author on an eleven-author review of 100 published studies on using brain-wave data to detect major depressive disorder.

Under Preparation

[14] Startup survival among newly public technology companies, testing whether patent-backed technology and artificial intelligence at initial public offering predict firm survival beyond financial measures alone, for *Technological Forecasting and Social Change*. First author.

Research Experience

ThinkNeuro 05/2026 – Present

Cognitive and Computational Neuroscience Program

A selective research training program in which students carry out and publish original neuroscience research under program mentorship.

Project Manager 08/2026 – Present

- Selected as the only high school student appointed to a project manager role in a cohort of more than 1,000 interns.
- Manage 5 student research groups, monitoring deliverables and progress, assisting with their research, and guiding them through submission to research conferences for presentation and publication.

Associate Research Intern 05/2026 – 08/2026

- Led a team of 11 undergraduate and high school researchers through a review of 100 published studies on brain-wave recordings and deep learning for major depressive disorder detection, and served as first author on the resulting abstract, presentation and electronic poster [13].
- First-authored the companion audit of data leakage in brain-wave depression-detection research, accepted for publication [5].

George Mason University & Alliance for a Green Revolution in Africa (AGRA) 04/2026 – Present

Research Intern, Aspiring Scientists Summer Internship Program Fairfax, VA

A national summer research program at George Mason University with an acceptance rate of approximately 9%, placing students directly into faculty-led research laboratories; placed with the Alliance for a Green Revolution in Africa (AGRA), a nonprofit that works to raise farm productivity and incomes across the African continent. Advised by Dr. Ron Mahabir, Dr. Maction Komwa and Dr. Olga Gkountouna at George Mason University, and by Dr. Wisdom Mgomezulu, a data scientist and researcher at AGRA.

- Built and audited the open database behind the first network study of development finance in African agriculture: 1,060 investor-to-deal records covering 389 transactions, 420 investors, 313 companies and 30 countries, assembled from 121 named sources [3].
- Wrote the data cleaning, company name-matching and audit software for that database, and personally verified 87.2% of records against an original source document.
- Mapped which investors fund deals alongside which others, across 420 investors and 202 shared transactions, and found that public development banks sit at the center of the network (6.04 times the brokerage centrality of other investors, $p < 0.0001$) yet rarely connect to private investors in practice, partnering with them at only 1.51 times chance against 4.92 times for private-to-private pairs [8].
- Shared findings with more than 20 African government officials; the work will be credited in the Alliance for a Green Revolution in Africa's 2027 Africa Food Systems Report, which is reviewed by policymakers and government officials across Africa.

University of North Texas 04/2026 – Present

Student Researcher, advised by Dr. Mohammed Aledhari Denton, TX

Research in applied machine learning within the Department of Computer Science and Engineering, focused on financial disclosure analysis and company-failure prediction.

- Built and validated a system that uses a large language model to score how evasively corporate executives answer analyst questions, across 3,350 exchanges from earnings calls filed with the United States Securities and Exchange Commission [7].
- Validated that system against two independent human raters and two outside datasets, reported the negative result honestly rather than tuning the method toward a favorable finding, and released the complete software publicly at a cost of $0.0038 per exchange scored.
- Built a company-failure prediction system over a 57,000 company-year dataset of 69 financial measures drawn entirely from free government filings, auditing the labels for contamination and tracking how accuracy shifted through the COVID-19 period [9].
- Extended that system into a startup feasibility model that reports failure risk and growth potential as two separate scores, so that a promising but risky venture is not averaged into the middle [14].

University of Texas Rio Grande Valley 03/2026 – Present
Student Researcher, Robust Intelligent Systems Laboratory Edinburg, TX
A university research laboratory studying whether machine learning systems remain trustworthy under realistic conditions. Advised by Dr. Sergei Chuprov.

- Ran 624 experiments across seven methods for training models on data split across many hospitals without sharing patient records, using 382,709 United States Centers for Disease Control and Prevention health survey records, and showed that a simple averaging baseline recovers 79 to 98% of the advantage each complex method claims [6].
- Built the complete experimental software for that study, along with a 181-check verification script that re-derives every number printed in the paper directly from the raw results.
- Created the Masking Gap, a measure of how far a single combined wellness score hides a person's worst individual health problem, and calibrated it against a random baseline on 8,353 National Health and Nutrition Examination Survey records [4].
- Generated 1,005 synthetic flood images from real traffic-camera frames, covering dangerous conditions that real camera footage rarely captures.
- Showed that a frozen I-JEPA encoder reaches 80.8% macro-recall while vision-language models fine-tuned on the same data transfer poorly to real photographs; under combined blur, darkness and noise the fine-tuned model missed every severe flood, while the frozen encoder held steady [1], [2].
- Presented this work at the IEEE Massachusetts Institute of Technology Undergraduate Research Technology Conference (October 2026), with two further papers accepted to the IEEE International Conference on Cognitive Machine Intelligence and the IEEE International Conference on Bioinformatics and Biomedicine (2026).

Projects

Posted / Completed

[1] **Question-Evasion Detection Framework**, **K. Chevuri**, A. Sheikh, M. Aledhari. University of North Texas, 2026. Code
Python. Public software that scores 3,350 corporate earnings-call exchanges for evasiveness at $0.0038 each, shipped with a single script that reproduces every statistic reported in paper [7], including its honestly reported null result.

[2] **African Agrifood Investment Dataset and Build Pipeline**, **K. Chevuri**, A. Poon et al. George Mason University, 2026. Code & Data
Python. Free 1,060-record database of agricultural investment covering 389 deals, 313 companies and 30 countries, released with the data cleaning, company name-matching and audit code that rebuilds it identically on every run. Underlies papers [3] and [8].

[3] **Synthetic Roadway Flood Dataset**, **K. Chevuri**, A. Peredo, S. Chuprov. University of Texas Rio Grande Valley, 2026. Dataset (Zenodo, CC BY 4.0) | Code
Python. 1,005 synthetic images generated from real traffic-camera frames, covering dangerous flood conditions that real footage rarely captures, released under a permanent DOI. Underlies papers [1] and [2].

[4] **Federated Learning Experiment Harness**, **K. Chevuri**, S. Chuprov. University of Texas Rio Grande Valley, 2026. Code
Python, PyTorch and Flower. Runs the full 624-condition experimental grid behind paper [6], paired with a 181-check script that verifies every reported number against the raw results.

[5] **Health Routine Analyzer**, **K. Chevuri**. Individual project, 2026.
Python, scikit-learn, FastAPI, Next.js and Supabase. Full-stack build: predicts a daily health score, then re-runs the model on modified inputs to quantify each recommendation ("sleep 1.5 more hours, score 47 to 55") instead of giving generic advice.

[6] **Energy Blackout Risk Monitoring Platform**, **K. Chevuri**, S. Rath, S. Jillellamudi. Prosper, TX, 2026.
Python. Built for the SkillsUSA Career Pathways Showcase, a national technical competition in which student teams design and defend an original engineering project. Led a three-person team from concept through national judging: first at district, first in Texas, and National Top 10.

[7] **3D Radar Detection System**, **K. Chevuri**, B. Kesanakurthy, S. Jillellamudi. Prosper, TX, 2025.
Python and Fusion 360. Led a three-person team building a three-dimensional radar-based detection system for the SkillsUSA Career Pathways Showcase: first at district and fifth in Texas.

[8] **Automated Dataset Cleaning and Analysis Platform**, **K. Chevuri** et al. Rayfield Systems, 2025.
Python, Pandas and scikit-learn. Anomaly detection, forecasting, clustering and visualization tool built as an intern team lead, presented to the chief executive officer and adopted by the company.

Planned

[9] **Personalized Learning Platform for Neurodivergent Students**, **K. Chevuri**. Individual project.
Will adapt instructional content and pacing to how an individual neurodivergent student learns best, drawing on failure-mode auditing methods from the research above to test where the platform is secretly underperforming for specific students rather than looking only at aggregate results.

[10] **Artificial Intelligence Startup Feasibility Platform**, **K. Chevuri**, M. Aledhari.
Will apply the government-filing models from papers [9] and [14] to score new business ventures on failure risk and growth potential as two separate outputs.

Honors & Awards

- **National Merit Commended Scholar**, August 2026. Awarded by the National Merit Scholarship Corporation to roughly the top 3% of the 1.3 million students who take the Preliminary SAT nationally.
- **SkillsUSA Career Pathways Showcase.** SkillsUSA is a national career and technical student organization; the Career Pathways Showcase judges original student-built engineering projects on technical merit and professional presentation. National Top 10, 2026, after first at district and first in Texas, for the Energy Blackout Risk Monitoring Platform; fifth in Texas, 2025, after first at district, for the 3D Radar Detection System.
- **DECA Entrepreneurship Series, Texas State Qualifier**, January 2026. DECA is an international student organization in marketing, finance, hospitality and management; the Entrepreneurship Series requires competitors to develop and defend an original business concept before industry judges.
- **Presidential Artificial Intelligence Challenge**, national competitor, January 2026.
- **Academic All-District Honors**, Prosper High School varsity soccer, 2023–24, 2024–25 and 2025–26, with **Honorable Mention All-District Honors** in 2025–26. District-wide recognition combining athletic performance with academic standing.
- **National Honor Society**, inducted Fall 2025. Membership by selection on scholarship, leadership, service and character.
- **Advanced Placement Scholar with Distinction**, July 2025. Awarded by the College Board for an average score of at least 3.5 across all Advanced Placement examinations taken, with scores of 3 or higher on five or more.
- **Gold President's Volunteer Service Award**, November 2024. The highest tier of a national award issued under the President's Council on Service and Civic Participation, earned for 220+ documented service hours with Hope4Spandana.

Leadership & Service

Hope4Spandana, Texas Chapter 03/2024 – Present
Co-Founder and President Prosper, TX
A nonprofit supporting people with physical and mental disabilities in India, funding housing, small-business and education projects. Co-founded and lead the Texas chapter, which raises funds locally and directs them to specific projects on the ground.

- Co-founded the Texas chapter and grew it from 6 active members to more than 100, building the officer structure, recruitment process and annual event calendar that sustain it.
- Raised more than $13,000 to date across more than 20 fundraisers, planning and running each from sponsorship through execution.
- Partnered with more than 10 businesses and nonprofits, including the North America Telugu Society, EkamUSA, Patel Brothers, Desi District and Swadeshi, converting those relationships into recurring sponsorship and donated venue space.
- Rebuilt the chapter's public outreach with printed booklets, rotating visuals and member training in audience engagement; the chapter has since been publicly recognized by three town mayors.
- Directed the funds raised to projects in India that now house 100 people in a shelter, support more than 20 storefronts owned by physically disabled entrepreneurs, and operate a school providing free education to 75 children with autism.
- Contributed more than 220 personal volunteer hours, earning the Gold President's Volunteer Service Award.

SkillsUSA, Prosper High School and Prosper Independent School District 08/2024 – Present
Chapter Officer and Competition Team Lead Prosper, TX
SkillsUSA is a national career and technical student organization serving more than 380,000 students, which runs district, state and national competitions in engineering and the skilled trades and trains members in professional and leadership skills.

Chapter Vice President 05/2026 – Present

- Grew the chapter by more than 70 members and led fundraising totaling roughly $5,000.

Public Relations Committee Head 09/2025 – Present

- Lead the committee responsible for how the chapter presents itself to students, families and local industry partners.
- Organized and ran more than 10 career-building and professional development workshops for members, covering interviewing, technical presentation and competition preparation.

District Reporter 05/2025 – Present

- Elected to run district-wide communications across Prosper Independent School District, covering chapter events, competition results and member recognition.

Competition Team Lead 08/2024 – Present

- Led a three-person team to first at district, first in Texas and National Top 10 in the 2026 Career Pathways Showcase with the Energy Blackout Risk Monitoring Platform.
- Led a three-person team to first at district and fifth in Texas in the 2025 Career Pathways Showcase with the 3D Radar Detection System.

Prosper High School Artificial Intelligence and Data Science Club 09/2025 – Present
Co-Founder and Vice President Prosper, TX
The school's first student organization devoted to data science and machine learning, founded to give students with no prior background a route into the field.

- Co-founded the club and built its meeting structure and introductory curriculum from nothing, writing lessons that assume no previous programming or statistics experience.
- Grew membership to more than 40 students and mentored four three-person teams through the DataJam and Congressional App Challenge competitions.

Prosper High School Solar Car Engineering Team 08/2023 – 04/2024
Computer-Aided Design Team Lead Prosper, TX
A student engineering team that designs and builds a solar-powered vehicle for competition.

- Led the computer-aided design subteam, producing the Fusion 360 models used in the team's vehicle design.

Work Experience & Entrepreneurship

Independent Soccer Coaching Business 06/2025 – 08/2025, 06/2026 – 08/2026
Founder and Head Coach Prosper, TX
A soccer coaching business founded and run independently, built around affordable access for families priced out of club training.

- Founded the business from nothing and handled every function personally: pricing, scheduling, marketing, parent communication and on-field coaching.
- Ran 10 or more revenue-generating camps with 10 or more players in each, and coached 5 or more players individually, across two summers.
- Set pricing deliberately below local club rates to keep training reachable for families in the community while still operating profitably.

Rayfield Systems 06/2025 – 07/2025
Machine Learning Intern and Team Lead
Placed at Rayfield Systems, a company operating in the energy and private market spaces, through the Harvard Undergraduate Venture Tech Summer Program, which admits roughly 15% of applicants.

- Led a 6-person team building a Python anomaly-detection and forecasting tool for energy data, using the Pandas and scikit-learn libraries.
- Personally implemented the anomaly detection, forecasting, clustering and visualization components.
- Pitched the finished platform to the chief executive officer; the company adopted it for internal use.

Athletics

Prosper High School Varsity Soccer 12/2023 – Present
Captain, Senior Season; Player Prosper, TX

- Selected as team captain for senior season.
- Earned Academic All-District Honors in three consecutive seasons, plus Honorable Mention All-District Honors in 2025–26.

Solar Soccer Club Academy 03/2024 – 08/2025
Player, Solar Academy 09B, Elite Clubs National League Dallas, TX
The Elite Clubs National League is one of the top two competitive levels of youth soccer in the United States. The Solar Soccer Club Academy is one of its member programs, training and competing year round on a national schedule.

- Competed through the 2023–24 and 2024–25 national tournament cycles while carrying a full research and academic course load.
- Team honors: Elite Clubs National League Texas Conference Champions and National Top 16 (2023–24); Dallas Cup Finalists and SuperCopa+ Champions (2024); Elite Clubs National League National Top 8 (2024–25); JFC Puma King Cup Champions (2025).

Technical Skills

Programming Languages & Tools: Python (primary research language, used on every project listed above), Java (Advanced Placement Computer Science A), R (certified), Unix shell, HTML, LaTeX; Git, Jupyter, RStudio, Linux and high-performance computing clusters (NVIDIA A100 graphics processing units), Fusion 360 computer-aided design, Excel, Overleaf

Programming Evidence: Six public repositories at github.com/kunalchevuri, covering the flood-robustness, synthetic-to-real, agrifood-dataset, federated-learning and evasion-detection papers. Each research repository ships its own verification script, including a 181-check script and a 278-check script that re-derive every number in the paper from the raw results and report any disagreement

Machine Learning & Deep Learning: PyTorch, Flower, Hugging Face Transformers, parameter-efficient fine-tuning (low-rank adaptation and its quantized variant), four-bit model quantization, scikit-learn, XGBoost, Pandas, NumPy, NetworkX, SHapley Additive exPlanations, Matplotlib

Federated Learning (training models across many institutions without sharing private records)**:** FedAvg, FedProx, SCAFFOLD, Ditto, q-FedAvg, Krum, trimmed mean, coordinate-wise median, cosine-similarity trust aggregation; Dirichlet and non-independent client partitioning; personalization evaluation

Vision & Language Models: SmolVLM2, Qwen2-VL, I-JEPA, SigLIP and vision transformer encoders, frozen encoder linear probing, fine-tuning and stability analysis, generative image editing for synthetic training data, large language models as automated evaluators (Claude, Llama 3.3)

Statistics & Experimental Design: Mann–Whitney, Wilcoxon and McNemar tests, Cohen's κ, bootstrap confidence intervals, permutation tests and configuration-model nulls, Louvain community detection, area under the receiver operating characteristic curve and operating-point analysis, power analysis, multiple-comparison correction

Data Sources: Securities and Exchange Commission EDGAR and XBRL filings; Centers for Disease Control and Prevention Behavioral Risk Factor Surveillance System, National Health and Nutrition Examination Survey and WONDER mortality database; Web of Science; traffic-camera imagery

Research Practice: reproducible build pipelines, automated result-verification scripts, public pre-registration of study designs, data-contamination auditing, inter-rater reliability protocols, and scientific writing and peer review response in IEEE format

Certifications, Memberships & Interests

Certifications: R Programming and Data Analytics, ThinkNeuro Cognitive and Computational Neuroscience Program (April 2026)

Memberships: National Honor Society, SkillsUSA, DECA

Interests: Reading, film study, music analysis, soccer